INCH POUND
MIL-PRF-3098/67D
27 August 1997
SUPERSEDING
MIL-C-3098/67C
28 June 1995

PERFORMANCE SPECIFICATION SHEET

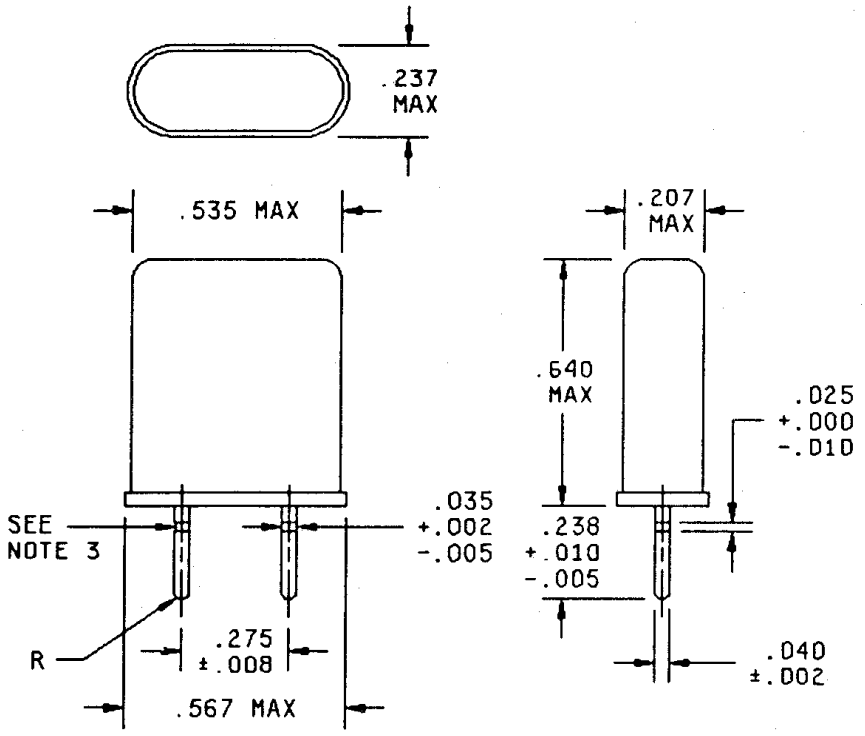
CRYSTAL UNIT, QUARTZ, CR89/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 2.12 MHz to 6.2 MHz; fundamental; noncontrolled; antiresonance.

Inches	mm
.002	.05
.005	.13
.008	.20
.010	.25
.025	.64
.035	.89
.040	1.02
.207	5.26
.237	6.02
.238	6.05
.275	6.99
.535	13.59
.567	14.40
.640	16.26



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Pin undercut may be omitted.
4. Marking shall be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR89/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.12 MHz to 6.2 MHz, inclusive.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Operating temperature range (noncontrolled): -55°C to +90°C, inclusive.

Frequency tolerance: ±50 ppm.

Equivalent resistance: See table I.

Antiresonance, load capacitance: 30.0 pF ±0.5 pF.

Capacitance, shunt: 7.0 pF, maximum.

Shock (specified pulse):

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202.

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

Frequency and equivalent resistance:

Frequency change permitted: ±5 ppm.

Equivalent resistance change permitted: ±15 percent.

TABLE I. Equivalent resistance.

Frequency range, inclusive MHz	Maximum resistance Ohms
2.12 to 2.25	500
2.25+ to 2.80	320
2.80+ to 3.41	250
3.41+ to 4.10	180
4.10+ to 5.10	120
5.10+ to 6.20	75

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Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC

Review activities:
Army - MI

Preparing activity:
Army - CR

Agent:
DLA - CC

(Project 5955-0697-40)

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